

PROJECT REPORT ON

TALK2MIND

A Multimodal AI-Based Mental Well-Being Assessment and
Support System

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1. Problem Statement

Mental health issues such as stress, anxiety, depression, sleep disorders, and emotional burnout have increased significantly in recent years. A large percentage of employees and students experience mental health challenges but remain unaware of their symptoms until the condition becomes severe. Traditional mental health assessments often require consultations with professionals, making early self-assessment difficult and inaccessible for many individuals.

Objective: Develop an AI-powered mental well-being assessment system that analyzes a person's emotional state through visual and speech-based interactions, enabling early identification of potential mental health concerns and providing personalized recommendations.

2. Methodology

Talk2Mind acts as an intelligent virtual mental wellness assistant. Users interact through a guided conversation session (~10–15 minutes), during which the system collects information from four sources:

- Facial Expression Analysis — computer-vision-derived emotion probabilities and micro-expression signals (eye contact ratio, blink rate, facial muscle tension, smile frequency, head movement variance).
- Speech Emotion Analysis — voice/audio-derived features (MFCC-style embeddings, pitch, energy, speech rate, pause ratio, jitter, shimmer).
- Questionnaire-Based Assessment — a PHQ-9 + GAD-7 inspired self-report questionnaire capturing depression- and anxiety-related indicators.
- Behavioral Feature Fusion — combining all three modalities into one representation.

Fused features are passed through a scoring engine (rule-based weighted sum, and a learned MLP regressor) to produce a Mental Well-Being Score (0–100) and a risk category (Good / Mild Concern / Moderate Concern / High Concern), which drives a personalized recommendation engine.

3. Dataset

Because collecting real facial video, audio, and questionnaire responses from human participants requires informed consent and ethics review, this phase uses a synthetic, label-consistent dataset generated by `data/generate_dataset.py`. A hidden per-user latent distress variable (Beta(2,5) distribution, reflecting a realistic population skew) drives correlated facial, speech, and questionnaire signals.

File	Rows	Columns	Description
questionnaire_responses.csv	2,000	19	PHQ-9 (9 items) + GAD-7 (7 items) responses + totals
facial_features.csv	2,000	30	Emotion probabilities, micro-expression signals, 16-dim visual embedding

File	Rows	Columns	Description
speech_features.csv	2,000	31	Emotion probabilities, acoustic features, 16-dim speech embedding
talk2mind_dataset.csv	2,000	83	Fused master table + Mental Well-Being Score + risk category

Risk category distribution in the generated sample (n=2,000, seed=42):

Category	Count
Mild Concern	1,117
Moderate Concern	547
Good	258
High Concern	78

4. Model Development

4.1 Facial Emotion Recognition

A RandomForestClassifier trained on the facial feature set (visual embedding + engineered signals) predicts the dominant facial emotion. In production, this would be replaced by a CNN backbone (ResNet/EfficientNet) fine-tuned on FER2013/AffectNet-style data, run frame-by-frame on webcam video and pooled at the session level.

4.2 Speech Emotion Recognition

A GradientBoostingClassifier trained on the speech feature set (MFCC-style embedding + acoustic features) predicts the dominant speech emotion. In production, this would be replaced by a BiLSTM/Transformer model trained on MFCCs extracted via librosa from raw audio (e.g. RAVDESS/CREMA-D-style training data).

4.3 Multimodal Feature Fusion

Facial and speech feature vectors, plus questionnaire scores, are concatenated and standardized, then fed into a small MLP regressor trained to predict the Mental Well-Being Score directly — letting the model learn data-driven modality weights rather than fixed hand-set ones.

4.4 Mental Well-Being Scoring Engine

- Rule-based: a transparent weighted sum of questionnaire, facial, and speech "distress proxies" (default weights: 50% questionnaire, 25% facial, 25% speech).
- Model-based: wraps the trained fusion MLP for a learned score.

4.5 Personalized Recommendation Engine

Given the risk category, specific high-scoring questionnaire items, and elevated facial/speech-distress signals, the engine returns a short, prioritized list of self-care suggestions. If a crisis indicator is detected, the engine always surfaces a crisis-support message ahead of generic tips.

5. Evaluation Results

Summary results on held-out synthetic test data (full run saved to data/evaluation_report.txt):

Component	Metric	Value
Facial Emotion Model	Accuracy	~0.99
Speech Emotion Model	Accuracy	~0.75
Fusion MLP (Well-Being Score)	R ² / MAE	~0.79 / ~4.8 (0–100 scale)
Risk Category (derived)	Accuracy	~0.79

These numbers reflect a synthetic, label-consistent dataset and must be re-validated on real, consented data before any real-world use.

6. Dashboard & User Interface

Two interfaces are included in the deliverables:

- A Streamlit dashboard (dashboard/app.py) where users complete the questionnaire, adjust sliders simulating facial/speech signals, and instantly see their score, risk category, and recommendations, plus a progress-over-time chart.
- A plain-language, standalone web check-in page (dashboard/talk2mind_checkin.html) that replaces clinical terminology (e.g. "PHQ-9", "facial muscle tension") with everyday language, aimed at non-technical users.

7. Limitations & Ethical Considerations

- Synthetic data: current model performance reflects a synthetic, label-consistent dataset and must be re-validated on real, consented data before any real-world use.
- Not a diagnostic tool: Talk2Mind is a self-awareness aid, not a replacement for professional clinical assessment.
- Bias & fairness: real deployment requires auditing across age, gender, accent, and cultural expression-of-emotion differences.
- Privacy: facial video and voice are sensitive biometric data; real deployment needs explicit consent, secure storage, encryption, and a clear data-retention policy.
- Crisis handling: any system like this needs a clear, tested escalation path to human support for high-risk/crisis cases — it should never be the only safety net.

8. Expected Deliverables — Status

Deliverable	Status
Facial Emotion Recognition Module	Complete (demo model; real CNN swap-in point documented)
Speech Emotion Analysis Module	Complete (demo model; real BiLSTM/Transformer swap-in point documented)
Mental Health Assessment Questionnaire	Complete
Multimodal Feature Fusion Model	Complete
Mental Well-Being Scoring Engine	Complete
Personalized Recommendation System	Complete
Source Code and Trained Models	Complete
GitHub Repository	Ready to push (source code included)
Project Documentation / Report	Complete (this document)
Final PPT Presentation	Complete (Talk2Mind_Presentation.pptx)
Final Project Demonstration	Run dashboard/app.py or open dashboard/talk2mind_checkin.html

9. Next Steps for a Production System

- Replace synthetic facial/speech features with real extraction (OpenCV/MediaPipe + a fine-tuned CNN for faces; librosa/parselmouth + a fine-tuned BiLSTM/Transformer for speech).
- Collect a small, real, consented pilot dataset to validate and re-calibrate the scoring weights.
- Add authentication, secure storage, and a database (e.g. PostgreSQL) for user history.
- Add clinician-in-the-loop review for High Concern / crisis-flag cases.
- Conduct a fairness/bias audit across age, gender, and accent groups.